ECS Starburst Dispensing Robot

Team #45
Section CB5

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I. ABSTRACT

We were tasked with creating a walking and dispensing robot that is meant to suit the needs of engineering career services and grainger students through delivering candy at the engineering library. Our design had to be able to dispense multiple candies at regular intervals and walk a respectable pace.

To achieve this we started by gathering an understanding of the stakeholders and their needs as well as familiarizing ourselves with the mechanism requirements. With the goals in mind, we began designing and iterating through dispenser mechanisms until we landed on our final slider-crank design.

The next task was to coordinate the dispenser with a walking robot body, powered by a single motor. This involved determining the proper gear ratios to meet the speed and dispensing interval requirements, along with determining the correct leg linkage design.

Once everything was together, the robot was able to perform within the project requirements and prove to be capable of satisfying stakeholder needs.

II. Project Goals and Design Specifications

As UIUC increases in student population, the need for deliveries to become automated is becoming ever more important. This project aims to generate a significant number of designs to solve various specific dispensing problems on campus. Our team, team#45, developed an Engineering Career Service (ECS) outreach candy dispensing robot. The motivation behind this is that ECS needs reliable and light-hearted ways to advertise their services to grainger students, and grainger students need career help! Using a candy-dispenser to drop branded candy can help both grainger students and ECS staff achieve their independent goals!

The main design specifications include the following:

1. Walking and dispensing occur using 1 motor and are timed to dispense a payload every 2m
2. The robot must walk at a pace of 4m/min or greater
3. The robot must fit within a box of l*w*h of 30*18*18 cm^3
4. Must hold 5 or more payloads
5. The robot must walk smoothly and without dropping the payload
6. Good craftsmanship and quality design choices should be evident

The specific requirements of our team included:

1. No use of wheels (we felt it wasn't in the spirit of the project)
2. Six legs for maximum centroid balancing area
3. Every design decision should be made with the least amount of friction in mind
    a. Bushings, delrin, metal on delrin contact, and double supported shafts

III. Detailed Design Discussion

The final design includes 1 motor, 1 set of batteries and battery holder. The Dispenser mechanism has a chute with an anti-jam spring-loaded door that can hold 5 payloads. The walker itself walks on 6 legs that are built to be as smooth as possible per design requirement 3. Overall, all constraints were followed and the project was a success!

Our dispenser mechanism utilizes a slider-crank four bar linkage. The slider acts as a carriage for the single candy that is being dispensed during the current cycle. The height of the slider is slightly less than the height of the candy, only allowing for a single candy to be dispensed during a cycle.

Once the slider is empty, it allows for another candy to fall into the carrier once it passes underneath the chute that holds the remaining candies. On the other end of the cycle, there is a hole in the base that the dispenser lies on. Once the slider passes over the hole, the candy then falls out, therefore dispensing the next candy.

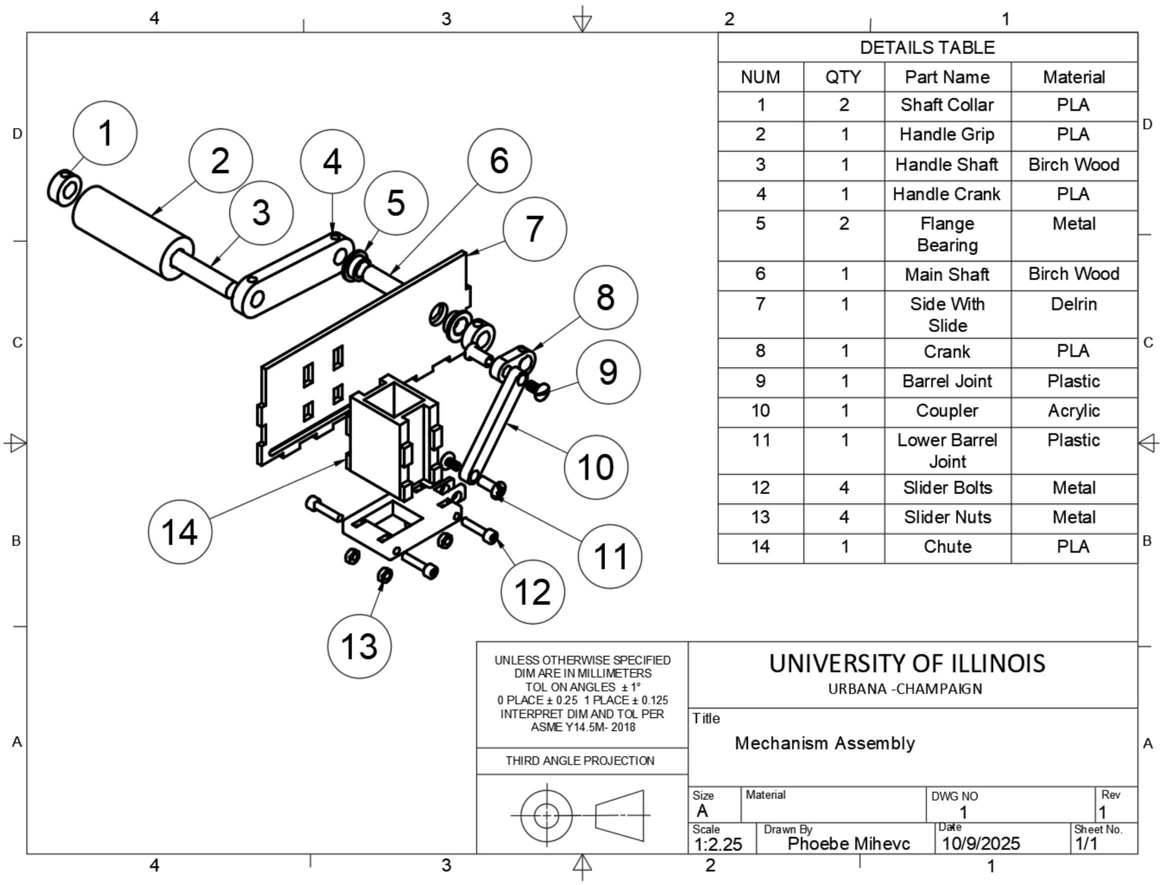


Figure 1. Dispenser Assembly Drawing

To ensure that the slider is secure and moves smoothly, there are 2 bolts on each side of the carriage which guide the slider's motion. These bolts enter through slots cut into delrin walls on either side of the dispenser and are threaded into 4 nuts that lie within the carriage. The head of the bolts hold against the walls and prevent the slider from moving laterally and the non-threaded part of the bolts rest on the delrin slots, ensuring low-friction interactions with the guiding slots.

The slider is connected with a plastic barrel fastener to the coupler which is connected at the other end to the crank using another barrel fastener. Then, the rotation of the crank is powered through a ⅜" dowell rod, which is secured tightly with the crank using a set screw that is threaded with a heat set insert on the crank.

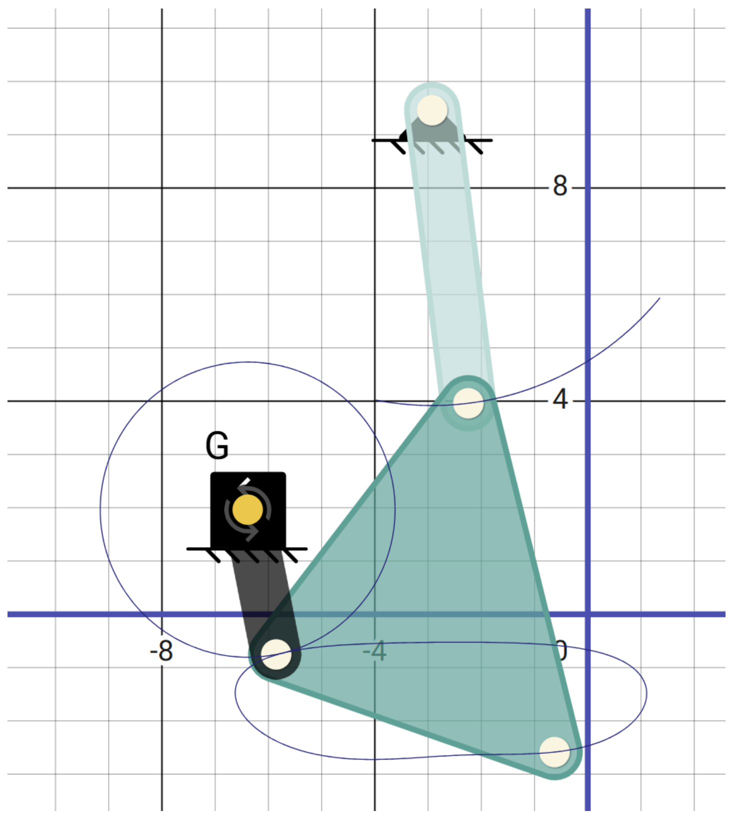


Figure 2. Leg Mechanism Gait Cycle

For the leg linkages, we sought a coupler path that consisted of a flat bottom portion of the cycle. We experimented using PMKS to find the proper design for this linkage and landed on the crank-rocker mechanism shown above (Figure 2.).

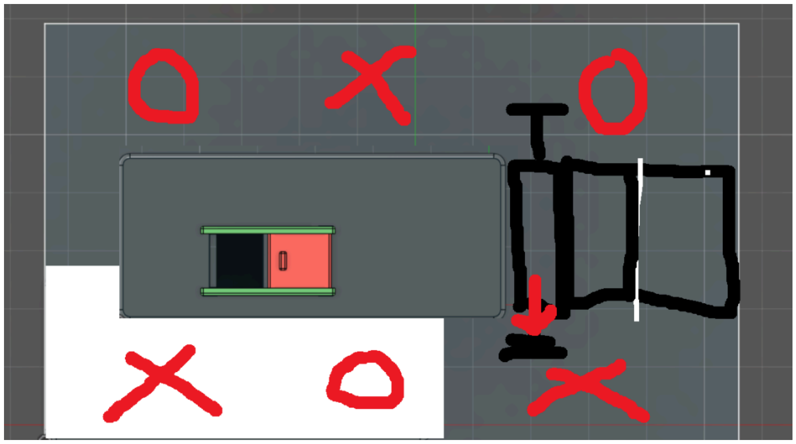


Figure 3. Leg Step/Gait Order

For the gait pattern, we opted for a 6-legged solution where we would have two groups of 3 legs, 180 degrees out of phase with each other. This means that while 3 legs are making contact with the ground, the other three would be in the returning phase of the gait cycle. The groupings consist of the two outermost legs of each side would be in phase with the middle leg of the other side. This results in a triangular contact patch, capturing a large patch at the center of the walker, ensuring stability due to the center of mass safely lying within the contact points.

Leading to the next step, which was finding the desired angular velocity of the leg crank, we needed to determine the distance the coupler path travels when it is making contact with the ground.

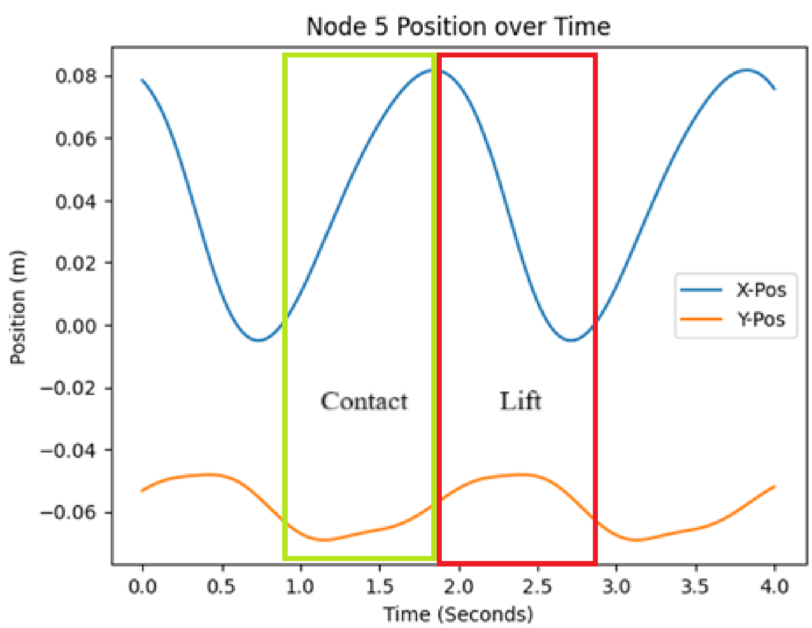


Figure 4. Foot Position vs. Time Plots

To determine the distance traveled per leg of the duty cycle, we used PVA analysis to identify when the legs are making contact and measured the displacement during that period which we found to be 7.87 cm or .0787m.

To coordinate the motion of the robot with the output of the motor, we began by finding the desired gear ratio between the motor and the legs. We did this by setting our desired walker speed to be 6m/min which is 2m/min above the requirement to account for a reduction in motor rpm due to load. Then we calculated the distance traveled by the walker per cycle which is double the distance per leg because both sets of legs make contact during a single cycle, therefore 0.1574m. Relating this to the desired walker speed, we end with a desired leg crank output of 34.48rpm.

The gear ratio between the legs and motor was then found by relating the motor output to the desired leg output. With a measured motor output of around 230 rpm, the desired ratio was 0.150.

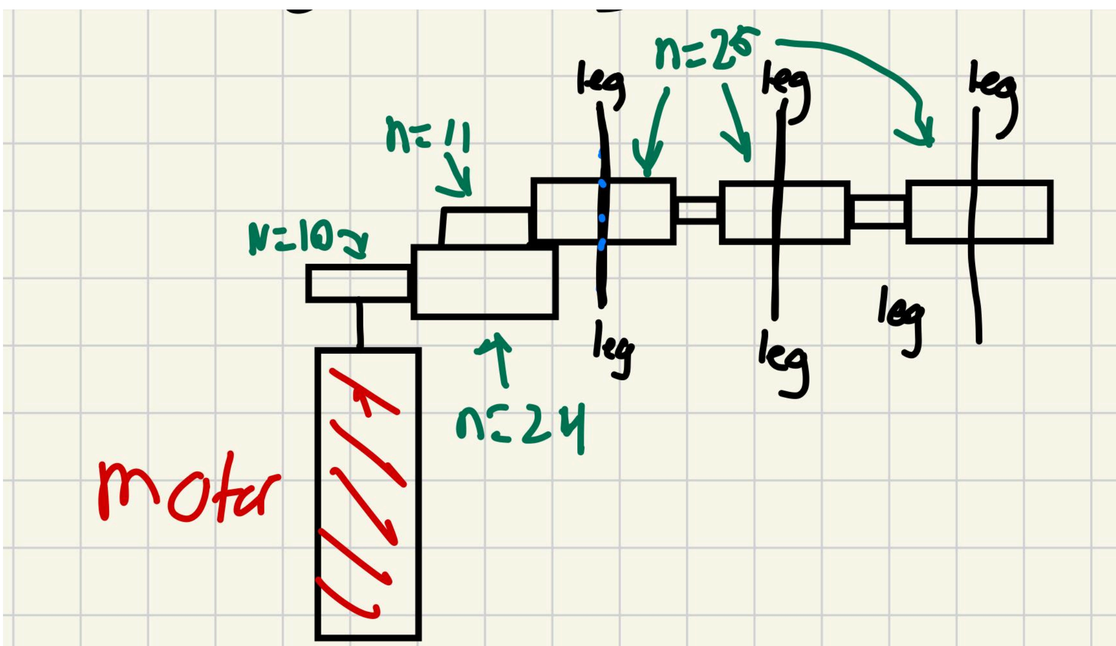


Figure 5. Leg Power Train Schematic

To achieve this gear reduction, we utilized the two-stage compound spur gear train shown above (Figure 5.). The gear ratio for this is 0.183, which is not exactly the target ratio, but it does not vary substantially enough to bring up any power transmission concerns. The ratio being higher also adds an extra buffer to make sure that the walker moves above the required speed of 4m/min. The leg gears are separated by idler gears which don’t affect the gear ratio, but instead make sure that all of the legs are moving toward the same direction.

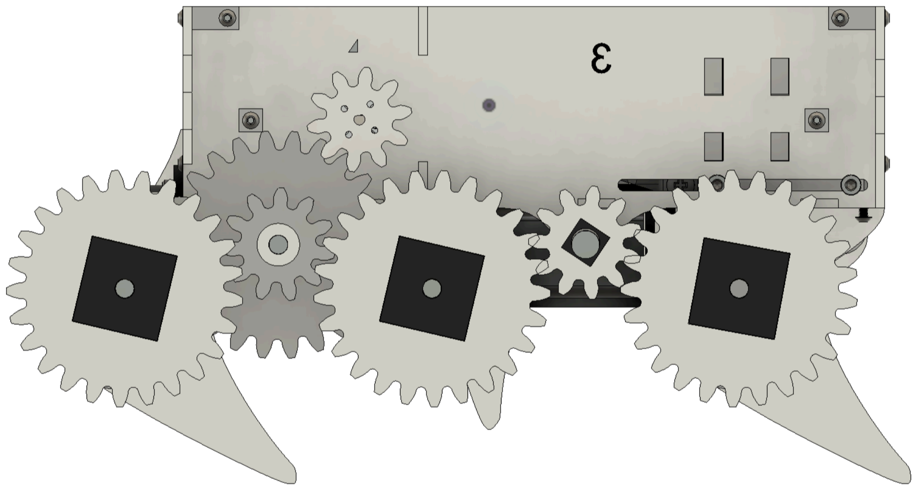


Figure 6. Exposed Gear Train

Figure 6. displays an exposed view of the gearing for the legs. Aside from the 24/11 tooth gear which was 3D printed using PLA. The remaining gears are made of delrin. Due to concerns with the laser cut gear being secure with the d shafts driving the legs, we designed 3D printed square keys to be press fitted within the gears, in the photo below (Figure 7.) it shows that these keys were designed to be secured to the d shaft using a set screw through a heat set insert.

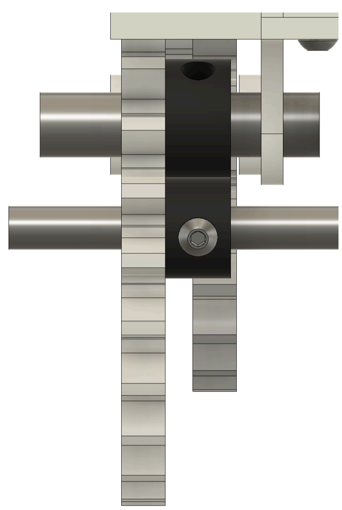


Figure 7. Spur Gear Secured with Key and Set Screw

Next, we were tasked with making sure that the dispenser timing is coordinated properly to make sure that the candies are dispensed at exactly every 2 meters. To do this, we then needed to calculate the gear ratio between legs and the dispenser crank.

The ratio was found by first determining the target dispensing cycle time. We calculated this by using the actual walker output and the desired distance between candy: $t_{disp} = \frac{d_{candy}}{V_{walker}}$, we measured the speed of the walker to be exactly 6m/min and with a desired distance of 2 meters, we calculated a dispensing time of 20 s.

Since exactly one candy dispenses per cycle, the desired dispensing output was 3 rpm. We related this to the leg crank output of 34.48 rpm and ended up with a desired gear ratio between the legs and the dispenser of 0.087.

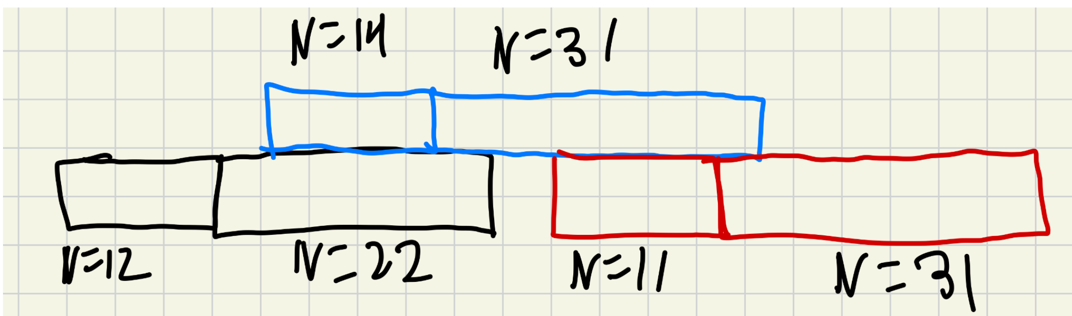


Figure 8. Dispenser Gear Train Schematic

To achieve this ratio, we designed a 3 stage compound spur gear train using the gear sizes shown above (Figure 8.).

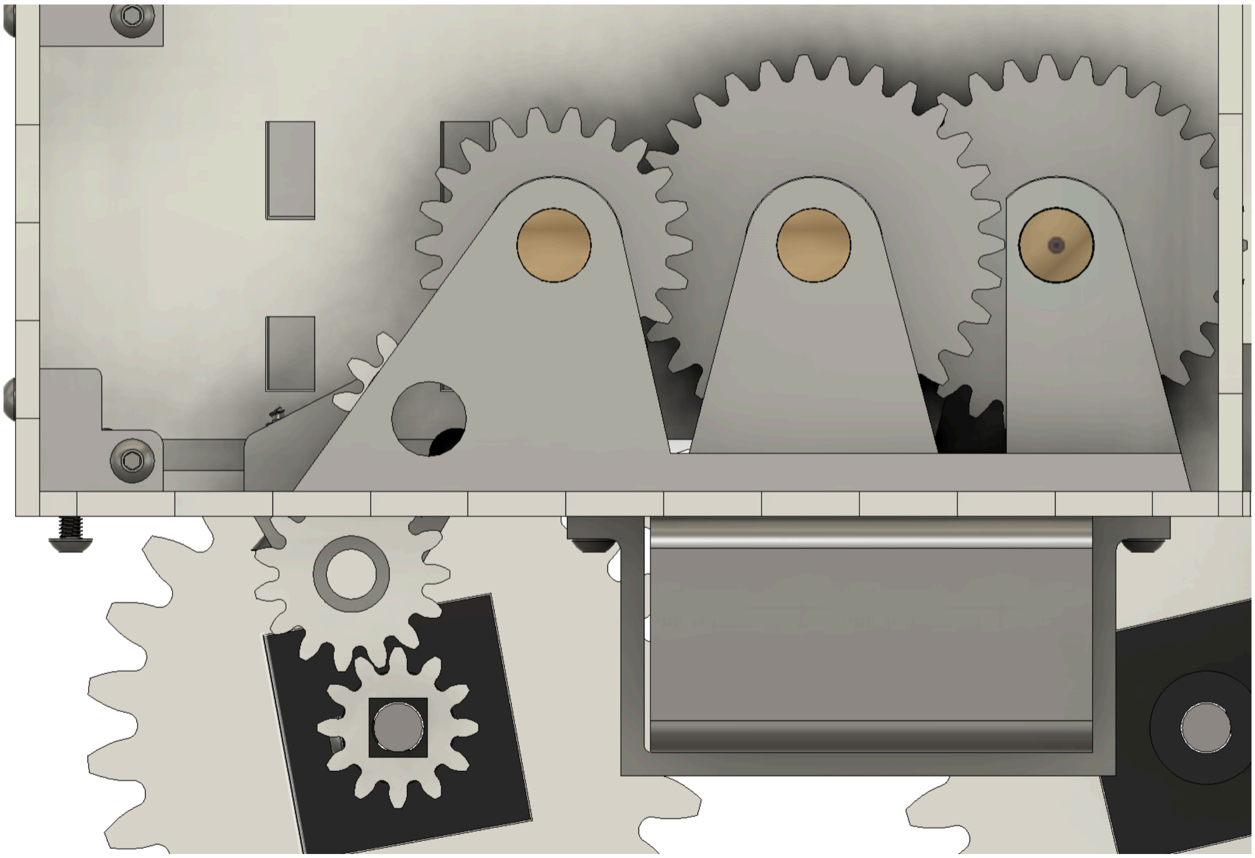


Figure 9. Dispenser Gear Train Exposed View

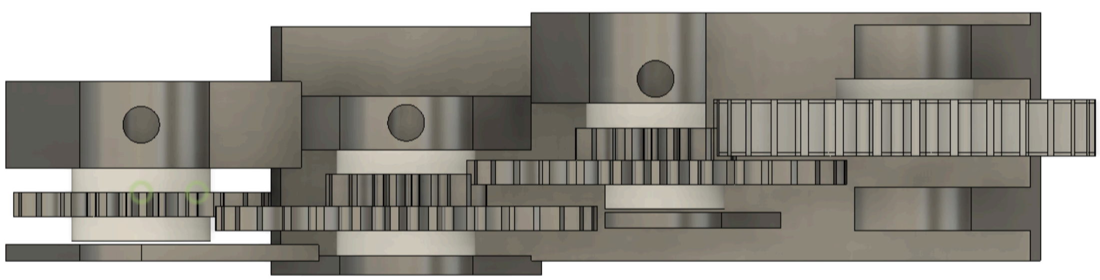


Figure 10. Dispenser Gear Train Top View

Above (Figure 9.) is an exposed view and a top view of the dispenser gearing (figure 10.). The compound gears were 3D printed and the others were laser cut out of delrin. The gears are held in the 3D printed gear holder and revolve around ⅜" dowel rods. The first gear is secured to the same d shaft that powers the legs, then it is connected to the rest of the gears using 2 15 tooth idler gears.

There was a plethora of iterations for each component of the robot, from the initial to end design. In the beginning, a recurring problem was inconsistent payload delivery and jamming. On occasion, the candy would load into the slider at an angle, and jam inside the chute. We increased the clearance on the inside of the slider walls, which helped, but did not remove the problem entirely. Next, we removed the front end of the slider (so that it was a fork-like shape), which forced the candy to dispense prematurely, so we moved the dispensing hole on the base forward, which led to an inconsistent delivery. We cut a hole into the back end of the chute, and tried various combinations of heights and chamfers until we found one that was successful, but again, did not completely solve the problem. Eventually, we found that the dispenser's slider was too tall: after the slider accepted a payload from the chute, the following candy would get caught on the slider instead of smoothly resting on the top of the slider. However, due to the dimension requirements imposed on the slider from the nuts and bolts used on the slider rails, the slider could not be made shorter. Instead, we moved the rails to be closer to the ground that the slider rides above.

The dispenser faced another problem when the last candy was to be dispensed; since there was no load on top of the candy to keep it horizontal until it dropped into the chute, it would sometimes fall into the slider vertically, and jam inside the chute as the dispenser moved forward for delivery. We tried chamfering the back of the slider hole to provide a gradual decline for the candy, which allowed the last candy to deliver, but caused the first four to jam. So, instead, we modified the chute to have a spring-loaded door that would remain closed unless enough force was applied to it from the inside. This way, if a vertical candy was sandwiched between the chute and the slider, it would be able to pass through. This eliminated the jamming issue, and gave assurance that the robot would not break from a stalled motor.

When we began to assemble the full walker for the first time, we encountered a few DFA problems. First, we couldn't connect the motor to the battery mount (which was beneath the main ground), so we cut a hole in the floor of the motor compartment, allowing the motor and wires to pass through. This also allowed us to remove the motor without having to fully disassemble the robot. Second, the wires were getting caught on the bottom axels, so we moved the battery mount to be closer to the motor hole. Third, the gears for the dispenser were hitting the back wall, so we cut a hole to provide more clearance.

Once the walker took its first steps, we found that it was consistently leaning to the left, with this behavior becoming more exaggerated the longer it walked. The leg joints were too loose, as the heatsets inside the legs and on the shaft collars bulged out, not allowing the joint pieces to sit completely flush against each other. Sanding down the parts helped, but did not resolve the issue. We reset the leg phase numerous times, attempting to make each phase perfectly aligned to each other, which did not help. A slow-motion video revealed that one of the front legs was hitting before the other, causing a "limp", so we offset the crank of that leg. This fixed the problem–but not for long lengths of time: the robot would eventually begin to turn. We realized that the screws that dug into the dowels on the legs were too short–sometimes not fully penetrating through the headsets. This was causing the joints to loosen greatly over time. After increasing the length of the screws, the loosening became negligible.

Another problem became apparent after walking–the current dispenser gearing was based on a different walker speed. The dispenser gear train was geared down too much, so we slightly changed the position and size of each gear, resulting in an average dispense interval of 1.98m (at the time of testing).

IV. Creative Theme

The walker is hand-painted to look like an octopus; one of the sides has a main body and all of the legs have tentacles. When in operation, it appears that the octopus is crawling across the floor as the tentacles move. This makes the robot fun and interesting to look at, but not overly distracting. Most of the parts (save for a few gears) are fully contained inside the main chassis, so attention is primarily drawn to the octopus design.

Aside from the octopus, “ECS” is painted in large print on the lid to bring attention to Engineering Career Services. The payload is intended to be ECS-branded candy that helps provide outreach to students.

V. Mechanism Performance

For the final event, the robot fit inside the size constraints, measuring approximately 16.5cm x 17.5cm x 30cm. The average distance between each dispensed payload was 1.95cm. The robot walked relatively straight, drifting about 2ft to the left over the 10m walked distance. The robot walked about 6m/min. The robot did not have any major problems, the only apparent issues being the slight drifting to the left and the 0.025% inaccuracy in the dispense intervals.

The dispensing interval differed from the calculated dispensing interval by $\pm$5.6cm, and the speed was roughly the same at 6.0m/min. We calculated the gearing to the dispenser via the speed of the robot without the dispenser. This is why speed matched so accurately. The dispensing on the other hand may have been off due to small slippage in the final leg phasing. This means that we moved less than 2m per dispensing event than planned.

VI. Reflection on Design Project

The machine performed very well. It fit nearly all the requirements except the budget where we spent slightly more than $100. There were challenges with walking straight, but we fixed them by changing the phase shift of the 6 legged design. If we were to iterate again we would elect to remove all metal shafts, bushings and many heat set inserts. It turned out that at this scale, the only requirement for smooth motion was double shaft support and the use of delrin. Additionally, it would have been nice to have a bit better grip on the ground to reduce slip.

Future teams should meet early and often, and work ahead of schedule. Things come up, and the best design is the one that is iterated the most. Overall, we were very happy with our team’s performance, and the final design quality.

VII. Appendix A (Drawings)

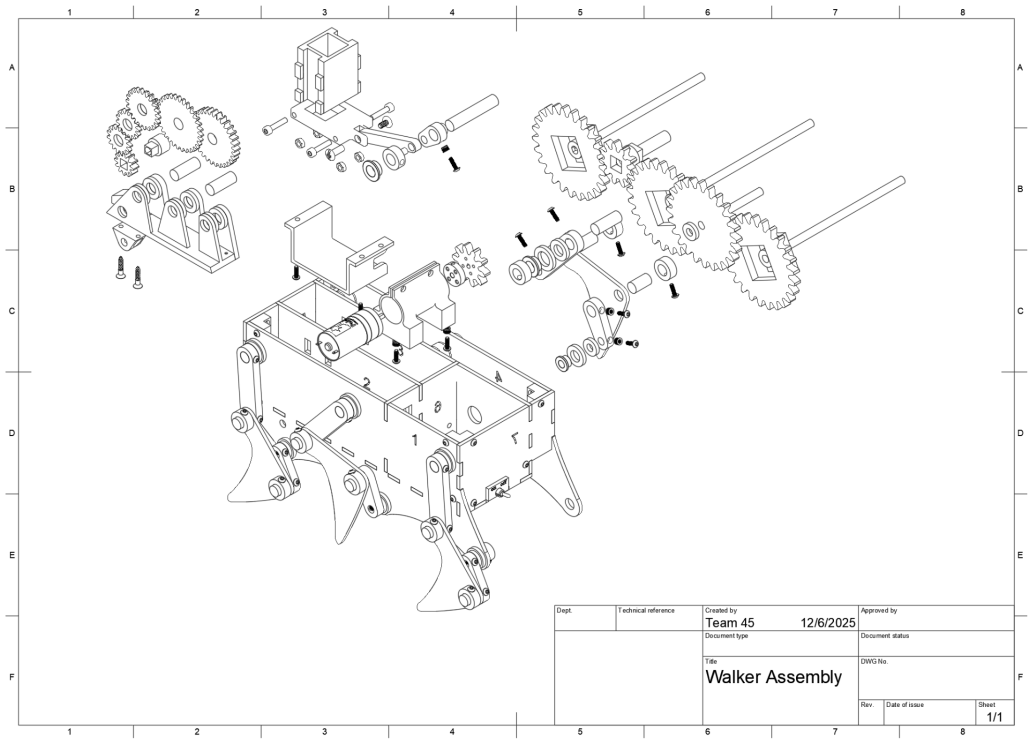


Figure 11. Walker Assembly Drawing

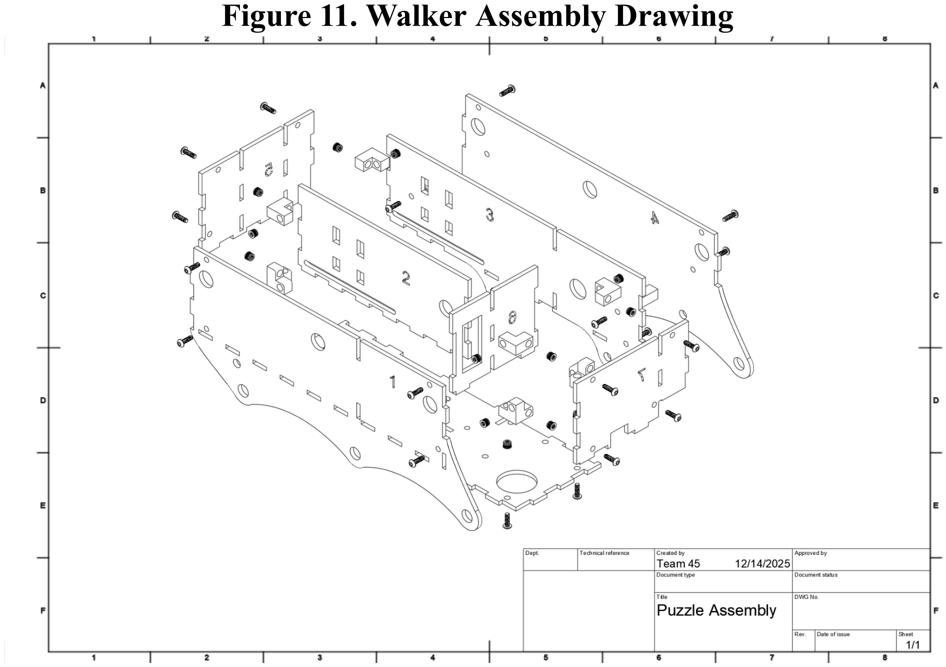


Figure 12. Puzzle Shell Assembly Drawing

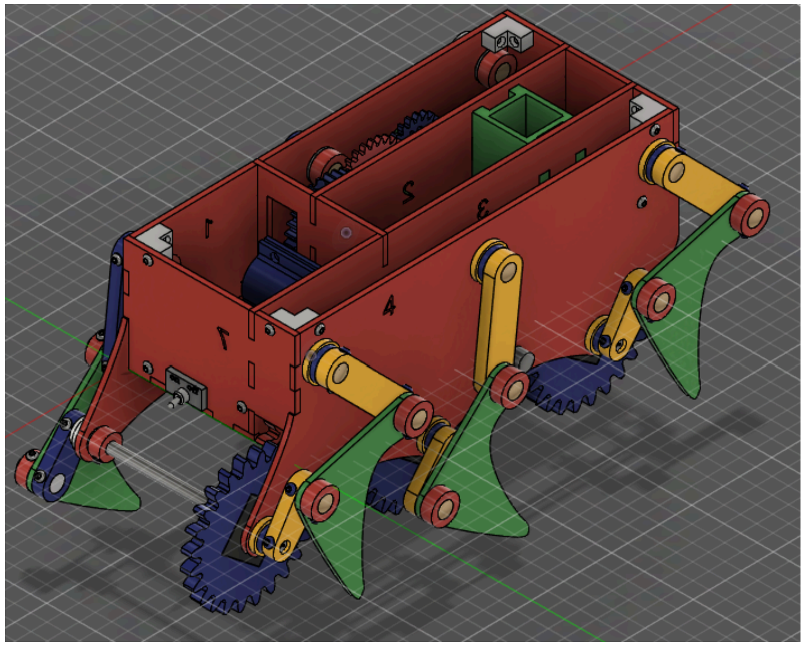


Figure 13: Full CAD view of final robot design

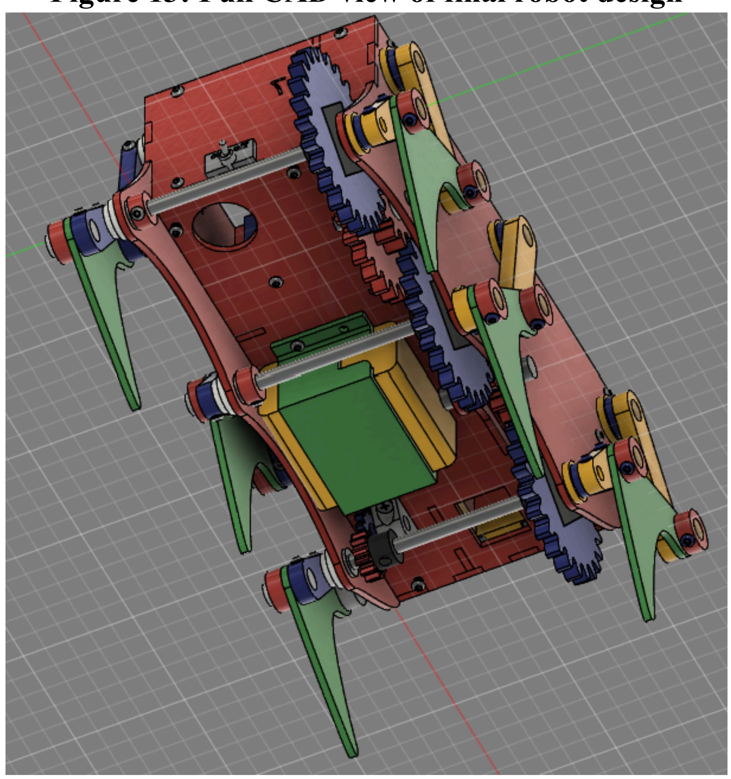


Figure 14: Underside view of final CAD design

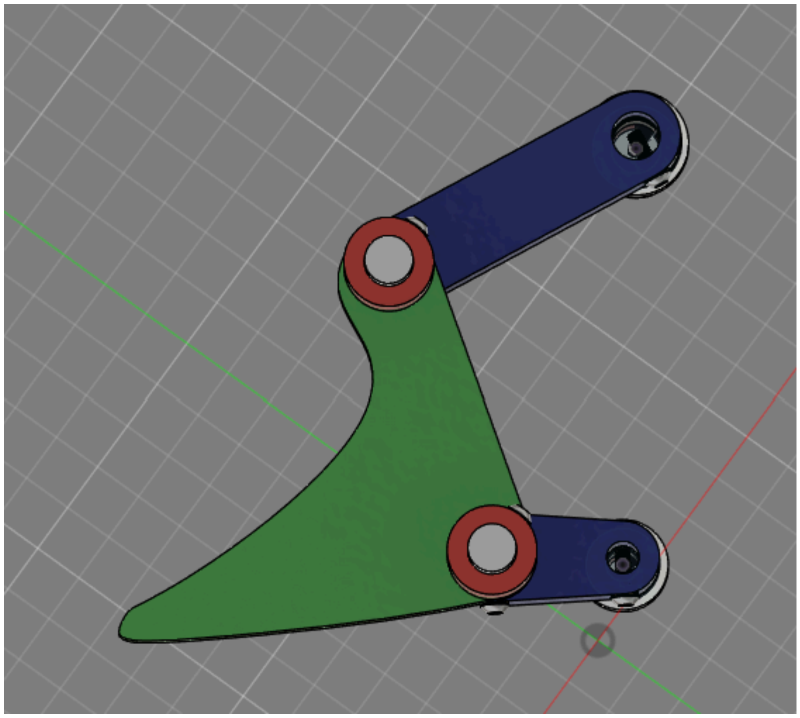


Figure 15: CAD view of leg design

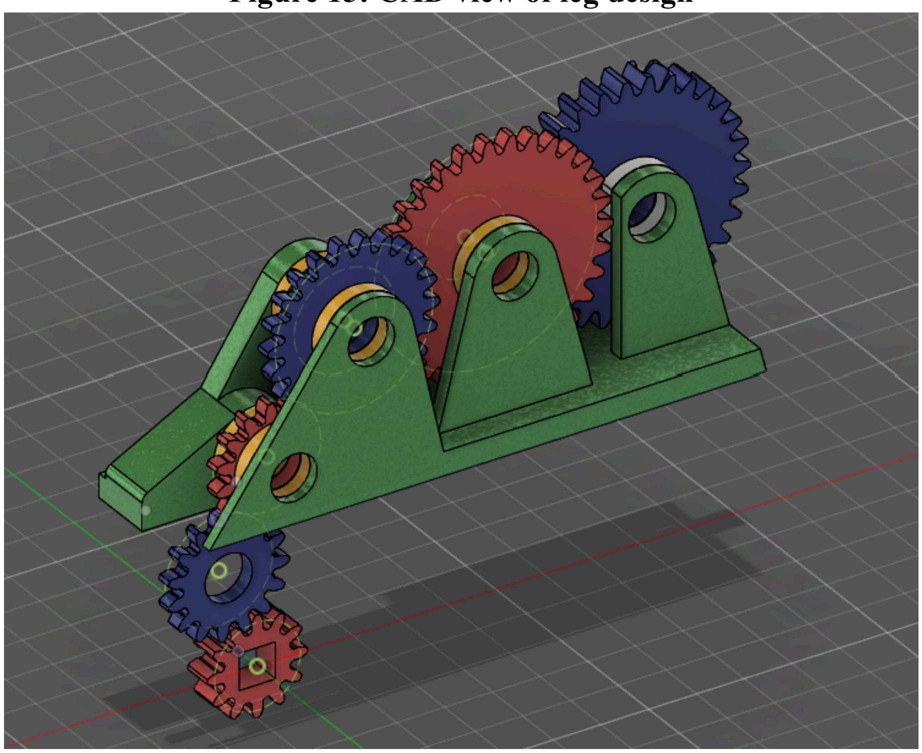


Figure 16: CAD view of dispenser gear train

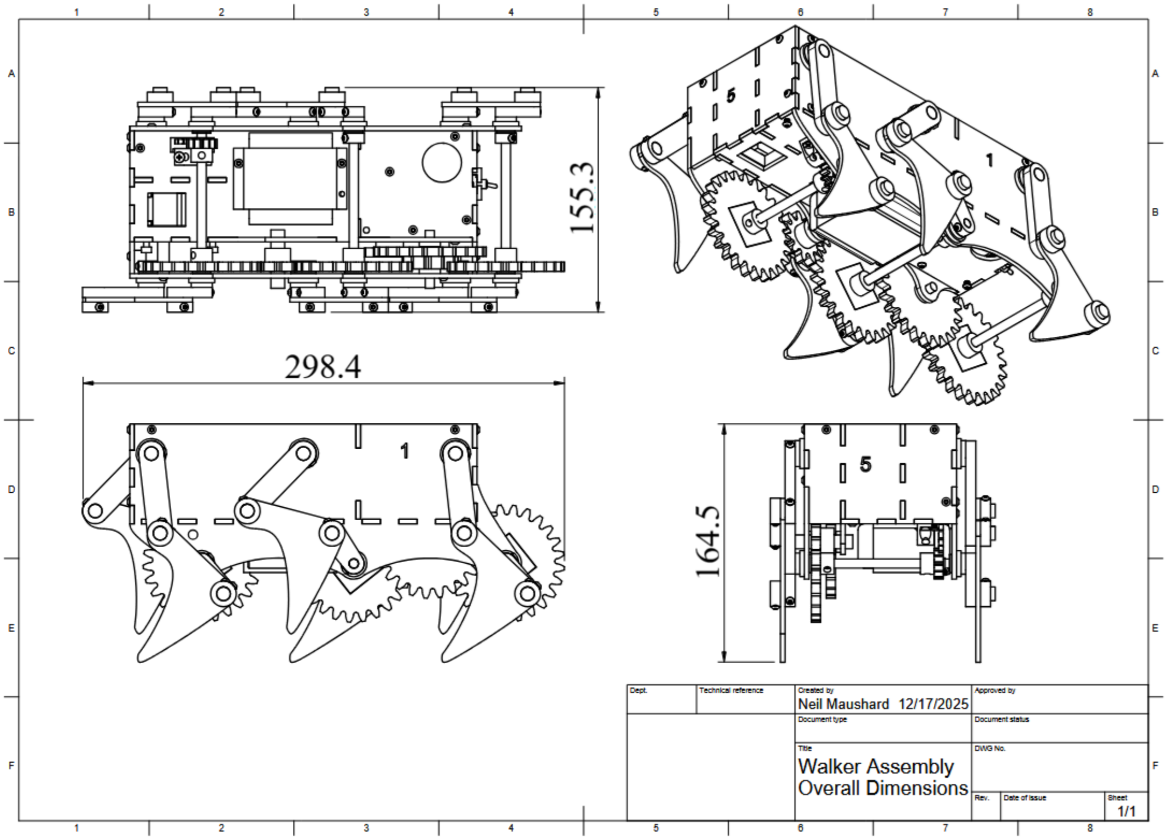


Figure 17: Overall dimensions of full assembly

Table 1. Bill Of Materials

Part Number	Part Name	Quantity	Material Name
1	1" Cut 3/8" Birch Rod	22	Birch Wood
2	Leg Rocker	6	PLA
3	Walker Foot	6	Delrin
4	Leg Crank	6	PLA
5	Chute	1	PLA
6	Dispenser Coupler	1	Acrylic
7	Slider Carriage	1	PLA
8	M4 Hex Nut	4	Steel
9	12mm M4 Screw	4	Steel
10	Barrel Fastener Screw	2	Plastic
11	Barrel Fastener Barrel	2	Plastic
12	2" Cut 3/8" Birch Rod	1	Birch Wood
13	Dispenser Crank	1	PLA
14	Dispenser Gear Holder	1	PLA
15	12 Tooth Gear Holder	1	PLA
17	Cutoff 3/8" Spacer	1	Delrin
18	Phillips Flat Head Screw	2	Steel
19	12 Tooth Dispenser Gear	1	Delrin
20	15 Tooth Dispenser Gear	2	Delrin
21	14/22 Tooth Dispenser Gear	1	PLA
22	11/31 Tooth Dispenser Gear	1	PLA
23	31 Tooth Dispenser Gear	1	PLA
24	D Shaft Key for 12 Tooth Gear	1	PLA
25	Motor Mount	1	PLA
26	10 Tooth Gear for Motor	1	Delrin
27	Motor Mounting Hub	1	Steel
28	Motor	1	Steel
29	24/11 Tooth Gear for Leg Powertrain	1	PLA
30	25 Tooth Gear for Leg Powertrain	3	Delrin
31	11 Tooth Gear for Leg Powertrain	1	Delrin
32	D Shaft Key for Large Gears	3	PLA
33	D Shaft Key for Small Gears	1	PLA
34	5" D Shaft for Legs	2	Steel
35	1 1/2" D Shaft	2	Steel
36	Battery Bracket	1	PLA
37	Battery Pack	1	Plastic
38	Switch Holder	1	PLA
39	Puzzle Shell Piece 1	1	Delrin
40	Puzzle Shell Piece 2	1	Delrin
41	Puzzle Shell Piece 3	1	Delrin
42	Puzzle Shell Piece 4	1	Delrin
43	Puzzle Shell Piece 5	1	Delrin
44	Puzzle Shell Piece 6	1	Delrin
45	Puzzle Shell Piece 7	1	Delrin
46	Puzzle Shell Piece 8	1	Delrin
47	Standoff Bracket	2	PLA
48	L Bracket	4	PLA
49	Corner Bracket	3	PLA
50	1/4" Shaft Collar	2	PLA
51	3/8" Shaft Collar	18	PLA
52	3/8" Spacer	17	Delrin
53	1/4" Spacer	14	Delrin
54	M3 Heat Set Insert	53	Brass
55	M3 Screw	53	Steel
56	1/4" Bushing	4	Brass
57	3/8" Bushing	7	Brass
58	5" Cut 3/8" Birch Rod for Legs	1	Birch Wood

VIII. Appendix B (Expense Report)

Our final expense tally left us slightly over-budget at $104.90, which we're happy with considering the quality of our project. The largest costs we incurred were due to raw material costs, totaling over $40 for PLA, sheets of delrin, and sheets of acrylic. We likely could have saved money by using more PLA than delrin, but this would have reduced the quality of our project since delrin's low friction surface helped us achieve smooth motion. We also spent a significant amount of money on heat set inserts, totalling $10.60 for heat sets alone. We used heat sets due to their strength and also because they allow us to fasten components together in a clean and neat way. We could have saved money by using nuts instead, but again, this would have led to a lower quality project. Our manufacturing costs remained pretty low, mostly due to us leaning heavily into laser cutting as opposed to 3D printing. Overall, despite being over-budget, we are happy with the choices we made when it came to component selection and budgeting due to the quality of our final product.

Table 2: Expenses from P1, excluding manufacturing costs

Lab section number, day, time:	BB6 (12:00-13:20 F)			Team Number and Names:	Team 45. Neil Maushard, Christopher Egly, Phoebe Mihevc, Roger Carstens	
Description	**Qty/$**	**Unit Price**	**Qty purchased**	**Source**	**Purpose**	**Total Cost**
3/8" dowel, 1' long	1/$0.5	$0.500	1	Innovation Studio	shafts	$0.50
Heat set insert	5/1$	$0.200	22	Innovation Studio	secure shell, secure components to shafts	$4.40
Flange Bearing, 3/8", SD, 1/4"	1/$	$1.000	2	Innovation Studio	smooth shaft rotation	$2.00
Plastic Barrel Fastener, 1/2" L	3 Sets/$1	$0.333	2	Innovation Studio	connect coupler to slider and crank	$0.67
HDRHS M3 x 10	6/$	$0.167	22	Innovation Studio	secure shell, secure components to shafts	$3.67
Hex Nut M4	30/$	$0.033	4	Innovation Studio	drop nut to secure M4 socket head to slider	$0.13
M4 Socket Head Screw 30 mm Long	100/13.63$	$0.140	4	Personal Purchase	provide sliding motion on rails	$0.56
PLA	1g/$0.03	$0.030	46.465	Innovation Studio	3D printing	$1.39
1/8" Delrin, 12x12"	1/$10	$5.000	1	Innovation Studio	laser cutting	$5.00
1/8" Acrylic, 12x12"	1/$5	$5.000	1	Innovation Studio	laser cutting	$5.00
Paint		$1.000	1	Personal Purchase	decoration	$1.00
Starburst	15/$2	$0.130	5	Personal Purchase	payload	$0.65
Manufacturing cost for laser	$0.001/cm	$0.001	39.15	Personal Purchase	back wall	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	88.10	Innovation Studio	bottom	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	84.90	Innovation Studio	crank support	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	37.89	Innovation Studio	front wall	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	60.24	Innovation Studio	side with hole	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	86.58	Innovation Studio	side with slide	$0.00
Manufacturing cost for laser	$0.001/cm	$0.001	18.53	Innovation Studio	coupler	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	2.41	Innovation Studio	carriage	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	4.39	Innovation Studio	chute	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.61	Innovation Studio	2x corner bracket	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.30	Innovation Studio	crank	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	5.41	Innovation Studio	grip	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	2.50	Innovation Studio	hand crank	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.31	Innovation Studio	lid door	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.13	Innovation Studio	2x lid rails	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	9.21	Innovation Studio	lid	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	11.38	Innovation Studio	base legs	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.68	Innovation Studio	3x shaft collar	$0.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.14	Innovation Studio	5x top bracket	$0.00
					P1 Total Spending:	**$24.98**

Table 3. Added expenses from P2

P2						
HDRHS M3 x 10	6/$	$0.167	31	Innovation Studio	secure shell, secure components to shafts	$5.18
heatset insert	5/1$	$0.200	31	Innovation Studio	secure shell, secure components to shafts	$6.20
Hex Nut M3	40/$	$0.025	2	Innovation Studio	secure motor mount closed	$0.05
Manufacturing cost for printing	$0.025 per cm^3	$0.025	14.4448	Innovation Studio	lid	$0.36
Manufacturing cost for printing	$0.025 per cm^3	$0.025	5.8412	Innovation Studio	motor mount	$0.15
Manufacturing cost for printing	$0.025 per cm^3	$0.025	3.4836	Innovation Studio	battery bracket	$0.09
Dispenser						
chute spring	15/$2.49	$0.166	1	Personal Purchase	keep chute closed	$0.17
Manufacturing cost for printing	$0.025 per cm^4	$0.025	4.95925	Innovation Studio	chute body	$0.12
Manufacturing cost for printing	$0.025 per cm^5	$0.025	0.5535	Innovation Studio	chute door	$0.01
Legs						
flange bearing part 79, 1/4"	1/$1	$1.000	4	Innovation Studio	mount inside wall for dshaft	$4.00
flange bearing part 82, 3/8"	1/$1	$1.000	5	Innovation Studio	mount inside wall for rocker	$5.00
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.3416	Innovation Studio	dhaft shaft collar x2	$0.01
Manufacturing cost for printing	$0.025 per cm^3	$0.025	3.405	Innovation Studio	3/8" shaft collar x15	$0.09
Manufacturing cost for printing	$0.025 per cm^3	$0.025	8.7767664	Innovation Studio	PLA coupler x6	$0.22
Manufacturing cost for printing	$0.025 per cm^4	$0.025	4.1028	Innovation Studio	PLA crank x6	$0.10
Manufacturing cost for laser	$0.001/cm	$0.001	19.9938	Innovation Studio	delrin foot x6	$0.02
Manufacturing cost for laser	$0.001/cm	$0.001	39.774	Innovation Studio	delrin spacer -1/4 x6	$0.04
Manufacturing cost for laser	$0.001/cm	$0.001	62.958	Innovation Studio	delrin spacer -1/2 x6	$0.06
Manufacturing cost for laser	$0.001/cm	$0.001	104.988	Innovation Studio	delrin spacer -3/8 x12	$0.10
walls and brackets						
Manufacturing cost for laser	$0.001/cm	$0.001	140.6	Innovation Studio	delrin wall 1	$0.14
Manufacturing cost for laser	$0.001/cm	$0.001	51.5	Innovation Studio	delrin wall 7	$0.05
Manufacturing cost for laser	$0.001/cm	$0.001	101.9	Innovation Studio	delrin wall 4	$0.10
Manufacturing cost for laser	$0.001/cm	$0.001	130.6	Innovation Studio	delrin wall 3	$0.13
Manufacturing cost for laser	$0.001/cm	$0.001	91.7	Innovation Studio	delrin wall 2	$0.09
Manufacturing cost for laser	$0.001/cm	$0.001	64.2	Innovation Studio	delrin wall 5	$0.06
Manufacturing cost for laser	$0.001/cm	$0.001	118.9	Innovation Studio	delrin wall 8	$0.12
Manufacturing cost for laser	$0.001/cm	$0.001	38.7	Innovation Studio	delrin wall 6	$0.04
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.9616	Innovation Studio	L bracket x4	$0.02
Manufacturing cost for printing	$0.025 per cm^4	$0.025	0.4596	Innovation Studio	standoff bracket x2	$0.01
Manufacturing cost for printing	$0.025 per cm^5	$0.025	0.3034	Innovation Studio	coner bracket x3	$0.01
Powertrain						
Manufacturing cost for laser	$0.001/cm	$0.001	171.9	Innovation Studio	delrin 25 tooth gear x3	$0.17
Manufacturing cost for laser	$0.001/cm	$0.001	24.8	Innovation Studio	delrin 11 tooth gear x1	$0.02
Manufacturing cost for laser	$0.001/cm	$0.001	22.2	Innovation Studio	delrin motor gear x1	$0.02
Manufacturing cost for printing	$0.025 per cm^4	$0.025	6.416	Innovation Studio	PLA 24/11 tooth gear x1	$0.16
Manufacturing cost for printing	$0.025 per cm^5	$0.025	1.354	Innovation Studio	PLA small dowel key x1	$0.03
Manufacturing cost for printing	$0.025 per cm^6	$0.025	4.0014	Innovation Studio	PLA 25 tooth gear key x3	$0.10
Shafts						
12" d shaft	1/$11	$11.000	1	Innovation Studio	main axle	$11.00
3/8" wood dowel	1/$0.50	$0.500	5	Innovation Studio	leg joints	$2.50
Material						
delrin 1/4" 12x12	1/$20	$20.000	0.5	Found	laser cutting	$10.00
delrin 1/8" 12x24	1/$10	$10.000	3	Innovation Studio	laser cutting	$30.00
PLA	1g/$0.03	$0.030	89.638	Innovation Studio	3D printing	$2.69
dispenser gear box						
part 6: Flathead #6, 0.75"	18/$	0.056	2	Innovation Studio	secure gear box bracket to gear box holder	$0.11
Manufacturing cost for printing	$0.025 per cm^3	$0.025	7.4916	Innovation Studio	gear box holder	$0.19
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.4188	Innovation Studio	gear box bracket	$0.01
Manufacturing cost for printing	$0.025 per cm^3	$0.025	2.2652	Innovation Studio	PLA 31 tooth gear	$0.01
Manufacturing cost for printing	$0.025 per cm^3	$0.025	2.3272	Innovation Studio	PLA 12/31 gear x2	$0.06
Manufacturing cost for printing	$0.025 per cm^3	$0.025	0.3812	Innovation Studio	PLA small dshaft key	$0.06
Manufacturing cost for laser	$0.001/cm	$0.001	0.229	Innovation Studio	delrin 10 tooth gear	$0.01
Manufacturing cost for laser	$0.001/cm	$0.001	21.431	Innovation Studio	delrin 20 tooth spur gear	$0.01
Manufacturing cost for laser	$0.001/cm	$0.001	15.848	Innovation Studio	delrin 15 tooth spur gear	$0.02
					P2 Total Spending:	**$79.92**
					P1 & P2 Total Spending:	**$104.90**

IX. Appendix C (Team Contributions)

Roger Carstens:

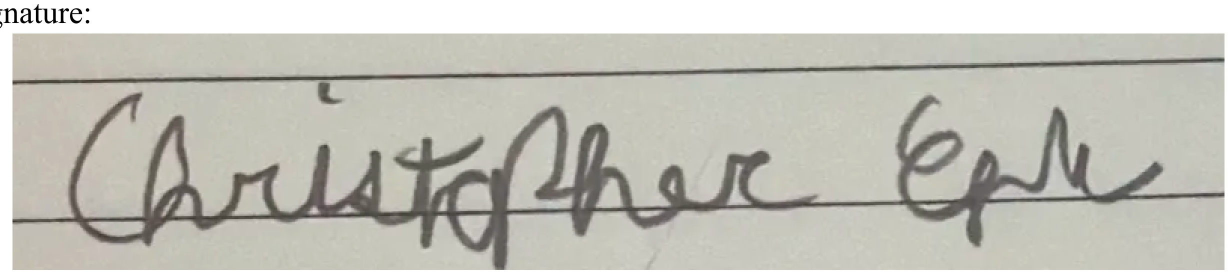
- Aided in developing themes along with ideating designs and mechanisms that met deliverable requirements.
- Calculated target gear ratios and designed and modeled gear trains
- I assert that all team members contributed significantly to report writing, CAD assembly, team meetings, and presentations

I, Roger Carstens, hereby agree that I followed all academic
integrity policies while producing this report. In addition to my original contributions, I have read through this entire report and certify that all materials, including the designs, are correct and not plagiarized.

Signature:
Date: 12/17/25

Christopher Egly:
- Developed significant portions of CAD:
- the crank-slider mechanism,
- P1 puzzle box
- P2 puzzle box
- Leg design
- Path synthesis
- I assert that ALL team members contributed significantly to report writing, CAD, assembly, team meetings, and presentations

I, Christopher Egly, hereby agree that I followed all academic
integrity policies while producing this report. In addition to my original contributions, I have read through this entire report and certify that all materials, including the designs, are correct and not plagiarized.
Signature:



Date:12/17/2025

Neil Maushard:
-Designed dispenser mechanism
-Contributed to general CAD design
-Path synthesis and leg design
-Assembly of physical components and sub assemblies
-Contributed to various aspects of report/presentation assignments

I assert that all team members contributed significantly to the project.

I, Neil Maushard, hereby agree that I followed all academic
integrity policies while producing this report. In addition to my original contributions, I have read through this entire report and certify that all materials, including the designs, are correct and not plagiarized.
Signature: *Neil Maushard*

Neil Maushard

Date:12/17/2025

Phoebe Mihevc

- Designed/iterated anti-jam chute mechanism, lid, and P1 legs
- Designed, decorated, and painted main chassis
- Assembled physical robot and made necessary design updates
- Gear key testing
- General CAD design
- I assert that ALL team members contributed significantly to report writing, CAD, assembly, team meetings, and presentations

I, Phoebe Mihevc, hereby agree that I followed all academic
integrity policies while producing this report. In addition to my original contributions, I have read through this entire report and certify that all materials, including the designs, are correct and not plagiarized.

Signature: Phoebe Mihevc
Date:12/17/2025

X. Appendix D

All members are okay with de-identified parts of the report being used as an example